Samkedemia
EDUCATION

ULTIMATE SOLUTION BOOK:
ALL EXERCISES & PAST EXAMS

Descriptive Statistics and Probability for Business

KU LEUVEN • BACHELOR OF BUSINESS ADMINISTRATION

VERSION: 4 NOV 2026

LET ME INTRODUCE MYSELF.

About Me

I'll make sure you pass this exam. I know exactly what it takes to get there.



BACKGROUND

- My name is **Jovan Samardzic**, I am 26 years old
- Got my **MSc in Statistics at KU Leuven**, graduated **Magna Cum Laude**
- Recipient of the **Science@Leuven Scholarship**

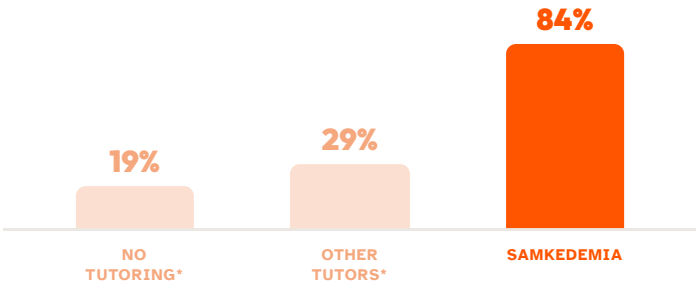
STYLE

- Founder of **Samkedemia**, helping students pass their exams
- Worked with **200+ university students** from all kinds of backgrounds
- **Plain language, not textbook theory** - just what actually makes sense

PROOF THAT IT WORKS.

Results

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Contents

1 Chapter D1 - Qualitative Variables **5**

Chapter D1 - Question 1 5

Chapter D1 - Question 2 8

Chapter D1 - Question 3 13

Chapter D1 - Question 4 18

Chapter D1 - Question 5 22

Chapter D1 - Question 6 27

2 Chapter D2 - Quantitative Variables **29**

3 Chapter D3 - Relationship Between Variables **30**

4 Chapter P1 - Probability **31**

5 Chapter P2 - Discrete Variables **32**

6 Chapter P3 - Continuous Variables **33**

7 Chapter P4 - Transformations of Random Variables **34**

8 Exam: 2020 June 1 **35**

9 Exam: 2020 June 2 **36**

10 Exam: 2021 June 1 **37**

11 Exam: 2021 June 2 **38**

12 Exam: 2021 August **39**

13 Exam: 2022 Trial **40**

14 Exam: 2022 Example **41**

15 Exam: 2024 June **42**

16 Exam: 2024 August **43**

17 Exam: 2025 Trial **44**

18 Exam: 2025 Example **45**

19 Exam: 2025 June **46**

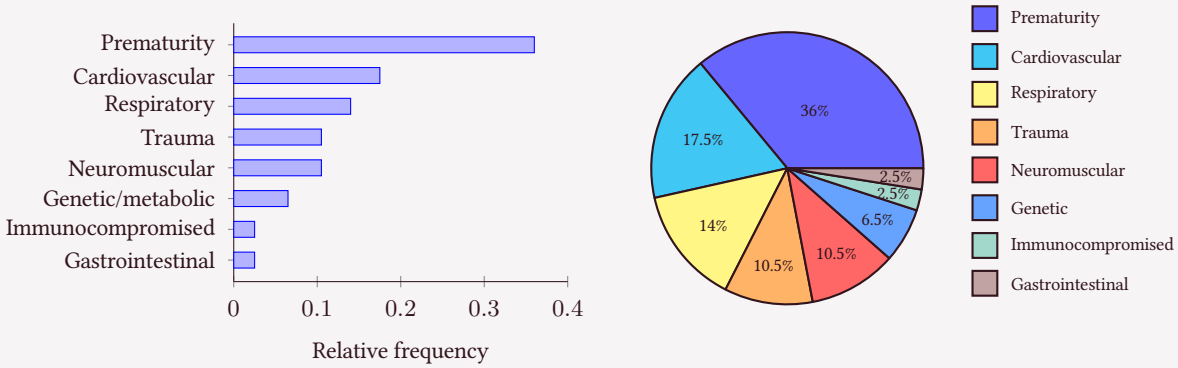
20 Exam: 2026 Trial **47**

21 Exam: 2026 Example **48**

Chapter D1 - Qualitative Variables

Question 1

The bar plot and the pie chart below show the distribution of pre-existing medical conditions of children involved in a study on the optimal duration of antibiotic use in treatment of tracheitis, which is an upper respiratory infection.



(a) What features are apparent in the bar plot but not in the pie chart?

(b) What features are apparent in the pie chart but not in the bar plot?

(c) Which graph would you prefer to use for displaying these categorical (=qualitative) data?

Tip: Bar Plot vs. Pie Chart

A **bar plot** (or a **needle graph**) uses bars to show how many times each category appears. Taller bars mean a category occurs more often. The bars can be arranged from largest to smallest to make comparisons easier.

A **pie chart** shows the same information but as slices of a circle. Bigger slices mean a category is more common. However, it can be harder to compare categories in a pie chart, especially when the slices are similar in size.

Solution 1: (a) What features are apparent in the bar plot but not in the pie chart?

The bar plot makes it easy to compare frequencies precisely, because all the bars sit on the same scale.

Also, when the bars are ordered by size, you can immediately see which conditions are most and least common.

It is also easier to tell apart categories with similar values.

Answer: Precise size comparisons and clear frequency rankings.

Solution 1: (b) What features are apparent in the pie chart but not in the bar plot?

There are no features that are apparent in the pie chart but not in the bar plot.

The bar plot already shows everything the pie chart shows. Because the bars use a relative-frequency scale, you can still see how each category relates to the whole sample, the same way you can in the pie chart.

Answer: No features - the bar plot already shows everything the pie chart shows.

Solution 1: (c) Which graph would you prefer to use for displaying these categorical data?

The bar plot is the better choice for this data because it is easier to read.

It lets you compare many medical conditions accurately and easily, even when their values are close together.

Pie charts, on the other hand, get harder to read once there are more than a few categories.

Answer: The bar plot, due to better readability and easier category comparison.

Question 2

In the table below, you find the hair colour for a sample of students.

blond	brown	black	blond	brown	brown	brown	blond	brown	black
black	blond	brown	black	black	black	brown	brown	brown	black
black	blond	black	brown	blond	black	blond	brown	brown	brown

(a) Set up a frequency table with absolute and relative frequencies for the variable *colour*.

(b) Find the mode of the variable *colour*.

(c) Set up a pie chart for the variable *colour*.

(d) Set up a bar chart for the variable *colour*.

Tip: Absolute and Relative Frequency

The **absolute frequency** tells us how many times a certain value appears in the data. It is just a simple count, denoted as $n_1, n_2, n_3, \ldots$ for different categories.

The **relative frequency** shows what share of all observations falls into that category. It is calculated as:

$$\text{relative frequency} = \frac{\text{absolute frequency}}{\text{total sample size}}$$

We denote relative frequencies as $\hat{p}_1, \hat{p}_2, \hat{p}_3, \ldots$. This makes it easy to compare categories, even if the total number of observations changes.

Solution 2: (a) Set up a frequency table with absolute and relative frequencies for the variable colour

Before computing anything, we set up the framework:

- **Random experiment:** Pick a random student and record their hair colour.
- **Random variable:**

  X hair colour of a random student

  X is qualitative, with $k = 3$ categories: $m_1 =$ black, $m_2 =$ blond, $m_3 =$ brown. It is nominal, since the categories have no natural order.

- **Given:**

  $n = 30$ sample size

  Their hair colours are denoted $x_1, x_2, \ldots, x_{30}$, using lowercase since these are observed values.

.....

Step 1: Count the absolute frequencies.

We count how many times each hair colour appears:

$$n_1 = 10, \quad n_2 = 7, \quad n_3 = 13$$

Step 2: Compute the relative frequencies.

$$\hat{p}_1 = \frac{10}{30} = 0.3333, \quad \hat{p}_2 = \frac{7}{30} = 0.2333, \quad \hat{p}_3 = \frac{13}{30} = 0.4333$$

Since colour is nominal, no cumulative relative frequency column is included.

Colour	Absolute Frequency	Relative Frequency
Black	10	0.3333 (or 33.33%)
Blond	7	0.2333 (or 23.33%)
Brown	13	0.4333 (or 43.33%)
Total	30	1 (or 100%)

Answer: The frequency table above.

Tip: Mode

The **mode** is the value that appears most often in the data.

For example, in the dataset: {1, 2, 2, 3, 3, 3, 4}, the mode is **3** because it appears the most times.

Solution 2: (b) Find the mode of the variable colour

The mode is the hair colour that appears most often. From the frequency table in part (a), brown appears 13 times, which is more than black (10 times) and blond (7 times).

Answer: Brown.

Solution 2: (c) Set up a pie chart for the variable colour

Step 1: Convert each relative frequency to a percentage of the circle.

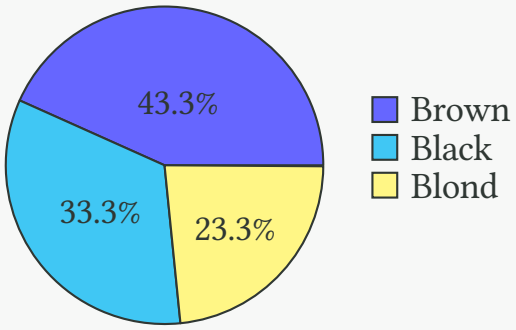
Using the relative frequencies from part (a): brown 43.3%, black 33.3%, blond 23.3%.

Step 2: Draw the circle and slices.

Divide a circle into three slices sized proportionally to these percentages, and label each slice

with its hair colour and percentage.

Answer: Pie chart shown below.



Solution 2: (d) Set up a bar chart for the variable colour

Step 1: Draw the axes.

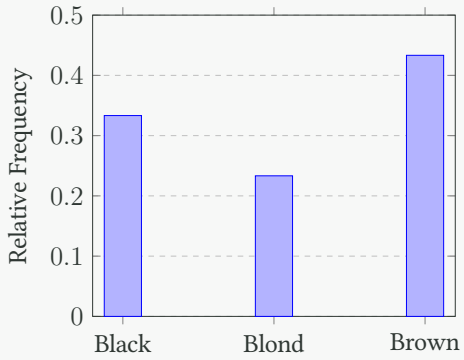
Draw a horizontal axis with one category per hair colour (black, blond, brown), and a vertical axis labeled "Relative frequency."

Step 2: Draw one bar per category.

Draw a bar for each category with height equal to its relative frequency from part (a).

Note that although bar charts are typically sorted in decreasing order, here we follow the category order introduced in part (a).

Answer: Bar chart shown below.



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Question 3

In the table below, you find T-shirt sizes for a sample of students, using the following coding:

1 = S, 2 = M, 3 = L, 4 = XL, 5 = XXL

4	2	4	3	1
2	3	1	2	1
3	3	3	3	2
4	4	1	1	2
3	2	3	1	3
1	2	1	1	5
1	4	3	2	2
3	3	2	1	2
3	1	4	5	4
2	3	3	1	3

(a) Set up a frequency table for the variable *Size of T-shirts* including absolute and relative frequencies.

(b) Add the cumulative absolute and relative frequencies in the table above.

(c) Create a needle graph and a pie chart for the T-shirt sizes.

Solution 3: (a) Set up a frequency table for the variable Size of T-shirts including absolute and relative frequencies

Before computing anything, we set up the framework:

- **Random experiment:** Pick a random student and record their T-shirt size.

- **Random variable:**

  X T-shirt size of a random student

  X is qualitative, with $k = 5$ categories: $m_1 =$ S, $m_2 =$ M, $m_3 =$ L, $m_4 =$ XL, $m_5 =$ XXL. It is ordinal, since the sizes have a natural order: S $<$ M $<$ L $<$ XL $<$ XXL.

- **Given:**

  $n = 50$ sample size

  Their T-shirt sizes are denoted $x_1, x_2, \ldots, x_{50}$, using lowercase since these are observed values.

. . . . . .

Step 1: Count the absolute frequencies.
We count how many times each size appears:

$$n_1 = 13, \quad n_2 = 12, \quad n_3 = 16, \quad n_4 = 7, \quad n_5 = 2$$

Step 2: Compute the relative frequencies.

$$\hat{p}_1 = \frac{13}{50} = 0.26, \quad \hat{p}_2 = \frac{12}{50} = 0.24, \quad \hat{p}_3 = \frac{16}{50} = 0.32$$

$$\hat{p}_4 = \frac{7}{50} = 0.14, \quad \hat{p}_5 = \frac{2}{50} = 0.04$$

Size	Code	Absolute Frequency	Relative Frequency
S	1	13	0.26
M	2	12	0.24
L	3	16	0.32
XL	4	7	0.14
XXL	5	2	0.04
Total	-	50	1 (or 100%)

Answer: The frequency table above.

Tip: Cumulative Frequency

The **cumulative absolute frequency** tells us how many observations fall in category i or an earlier one. It has no dedicated symbol in this course - it is just a running sum of the absolute frequencies:

$$n_1 + n_2 + \cdots + n_i$$

It is a minor, secondary column; the cumulative relative frequency below is the one that actually matters.

The **cumulative relative frequency** $\hat{F}_i$ does the same for relative frequencies:

$$\hat{F}_i = \hat{p}_1 + \hat{p}_2 + \cdots + \hat{p}_i$$

Both are only meaningful for ordinal variables, because they rely on the categories having a natural order. For example, in T-shirt sizes (S, M, L, XL, XXL), the cumulative frequency at L tells us how many students wear size L or smaller. For a nominal variable, such as hair colour, cumulative frequencies do not make sense, since the categories have no natural order.

Solution 3: (b) Add the cumulative absolute and relative frequencies in the table above

Step 1: Compute the cumulative absolute frequencies.

There is no dedicated symbol for this in the course - we just keep a running sum of the absolute frequencies, category by category:

$$13, \quad 13 + 12 = 25, \quad 25 + 16 = 41$$

$$41 + 7 = 48, \quad 48 + 2 = 50$$

Step 2: Compute the cumulative relative frequencies.

We add up the relative frequencies category by category:

$$\hat{F}_1 = \hat{p}_1 = 0.26$$

$$\hat{F}_2 = \hat{F}_1 + \hat{p}_2 = 0.26 + 0.24 = 0.50$$

$$\hat{F}_3 = \hat{F}_2 + \hat{p}_3 = 0.50 + 0.32 = 0.82$$

$$\hat{F}_4 = \hat{F}_3 + \hat{p}_4 = 0.82 + 0.14 = 0.96$$

$$\hat{F}_5 = \hat{F}_4 + \hat{p}_5 = 0.96 + 0.04 = 1.00$$

Size	Code	Abs. Freq.	Rel. Freq.	Cumul. Abs. Freq.	Cumul. Rel. Freq.
S	1	13	0.26	13	0.26
M	2	12	0.24	25	0.50
L	3	16	0.32	41	0.82
XL	4	7	0.14	48	0.96
XXL	5	2	0.04	50	1.00
Total	-	50	1 (or 100%)	-	-

Answer: The updated frequency table above.

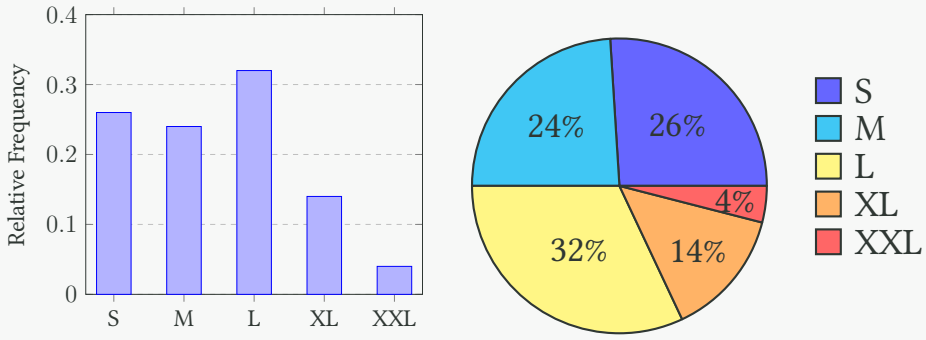
Tip: Needle Graph

A **needle graph** is another name for a bar chart.

Solution 3: (c) Create a needle graph and a pie chart for the T-shirt sizes

We plot the relative frequencies from part (a): one bar per size for the needle graph, one slice per size for the pie chart.

Answer: Needle graph and pie chart shown below.



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Question 4

In a frequency table, absolute frequencies are provided for the different outcomes for the variable *continent* for a sample of BBA students. This variable is a qualitative variable indicating the continent for a random BBA student.

Continent m_i	n_i	$\hat{p}_i$
Europe	159	
Asia	54	
N-America	11	
S-America	5	
Africa	26	
Oceania	1	

(a) Determine the sample size.

(b) Add a column with the relative frequencies in the table.

(c) Determine the mode.

(d) Draw a needle graph with absolute frequencies.

(e) Draw a pie chart with relative frequencies.

Solution 4: (a) Determine the sample size

Before computing anything, we set up the framework:

- **Random experiment:** Pick a random BBA student and record their continent.
- **Random variable:**

X continent of a random BBA student

X is qualitative, with $k = 6$ categories: Europe, Asia, N-America, S-America, Africa, Oceania. It is nominal, since continents have no natural order.

- **Given:** The absolute frequencies n_i for each continent, as shown in the table above.

The sample size n is the sum of all absolute frequencies:

$$n = 159 + 54 + 11 + 5 + 26 + 1 = 256$$

Answer: The sample size is $n = 256$.

Solution 4: (b) Add a column with the relative frequencies in the table

We compute the relative frequencies:

$$\hat{p}_1 = \frac{159}{256} = 0.6211, \quad \hat{p}_2 = \frac{54}{256} = 0.2109, \quad \hat{p}_3 = \frac{11}{256} = 0.0430$$

$$\hat{p}_4 = \frac{5}{256} = 0.0195, \quad \hat{p}_5 = \frac{26}{256} = 0.1016, \quad \hat{p}_6 = \frac{1}{256} = 0.0039$$

Continent m_i	n_i	$\hat{p}_i$
Europe	159	0.6211
Asia	54	0.2109
N-America	11	0.0430
S-America	5	0.0195
Africa	26	0.1016
Oceania	1	0.0039
Total	256	1 (or 100%)

Answer: The completed frequency table above.

Solution 4: (c) Determine the mode

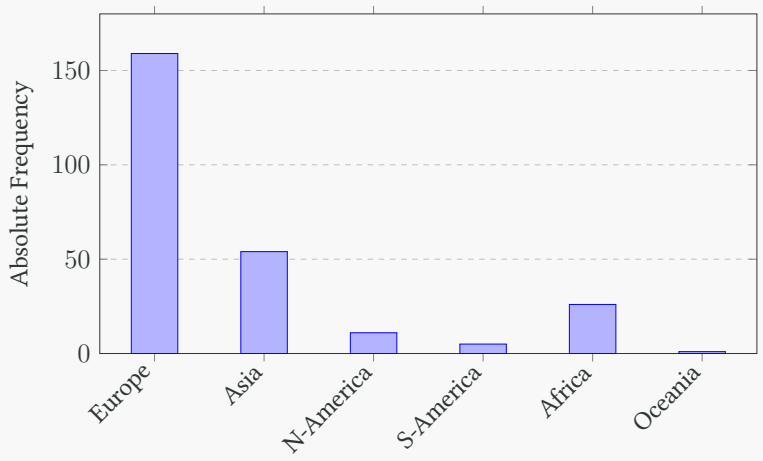
The mode is the continent with the highest absolute frequency. From the table, Europe has the highest count, 159.

Answer: Europe.

Solution 4: (d) Draw a needle graph with absolute frequencies

We plot the absolute frequencies from the table: one bar per continent.

Answer: Needle graph shown below.

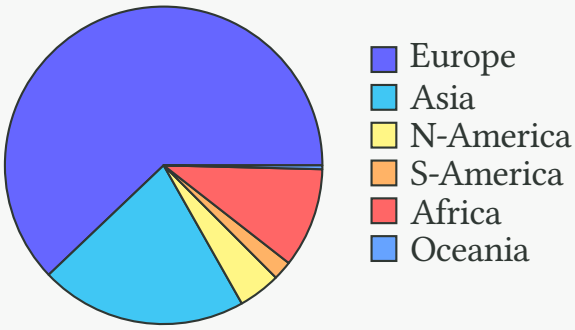


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Solution 4: (e) Draw a pie chart with relative frequencies

We plot the relative frequencies from part (b): one slice per continent.

Answer: Pie chart shown below.



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Question 5

For a sample of 519 members of academic staff in the USA, the rank is measured. For the ranks the following codes are used:

1 = Assistant Professor, 2 = Associate Professor
3 = Full Professor, 4 = Distinguished Professor

Code	Abs. Freq.	Rel. Freq.	Cumul. Abs. Freq.	Cumul. Rel. Freq.
1		0.35453		
2		0.34104		
3		0.25241		
4		0.05202		

(a) Complete the frequency table by determining the absolute frequencies (a.f.), the cumulative absolute frequencies (c.a.f.) and the cumulative relative frequencies (c.r.f.).

(b) Determine the mode.

(c) Draw a needle graph with relative frequencies.

(d) Draw a 100% stacked column chart.

Tip: Calculating Absolute Frequency

The **absolute frequency** for a category is found by multiplying its relative frequency by the total sample size:

$$n_i = \hat{p}_i \cdot n$$

Since a count must be a whole number, round n_i to the nearest integer. The sum of all absolute frequencies must always equal the total sample size - if rounding makes the sum slightly off, adjust one category up or down by 1 to fix it.

Solution 5: (a) Complete the frequency table

Before computing anything, we set up the framework:

- **Random experiment:** Pick a random staff member and record their rank.
- **Random variable:**

  X rank of a random staff member

  X is qualitative, with $k = 4$ categories: $m_1 =$ Assistant Professor, $m_2 =$ Associate Professor, $m_3 =$ Full Professor, $m_4 =$ Distinguished Professor. It is ordinal, since the ranks have a natural order: Assistant $<$ Associate $<$ Full $<$ Distinguished.

- **Given:**

  $n = 519$ sample size

  The relative frequencies $\hat{p}_i$ for each rank are shown in the table above.

Step 1: Compute the absolute frequencies.

$$n_1 = 0.35453 \cdot 519 = 183.9999 \approx 184, \quad n_2 = 0.34104 \cdot 519 = 176.9998 \approx 177$$

$$n_3 = 0.25241 \cdot 519 = 130.9998 \approx 131, \quad n_4 = 0.05202 \cdot 519 = 26.9994 \approx 27$$

Checking the total:

$$184 + 177 + 131 + 27 = 519$$

This matches the sample size, so the rounding is correct.

Step 2: Compute the cumulative absolute frequencies.

There is no dedicated symbol for this in the course - we just keep a running sum of the absolute

frequencies, category by category:

$$184, \quad 184 + 177 = 361$$

$$361 + 131 = 492, \quad 492 + 27 = 519$$

Step 3: Compute the cumulative relative frequencies.

$$\hat{F}_1 = 0.35453$$

$$\hat{F}_2 = \hat{F}_1 + 0.34104 = 0.69557$$

$$\hat{F}_3 = \hat{F}_2 + 0.25241 = 0.94798$$

$$\hat{F}_4 = \hat{F}_3 + 0.05202 = 1.00000$$

Code	Abs. Freq.	Rel. Freq.	Cumul. Abs. Freq.	Cumul. Rel. Freq.
1	184	0.35453	184	0.35453
2	177	0.34104	361	0.69557
3	131	0.25241	492	0.94798
4	27	0.05202	519	1.00000
Total	519	1 (or 100%)	-	-

Answer: The completed frequency table above.

Solution 5: (b) Determine the mode

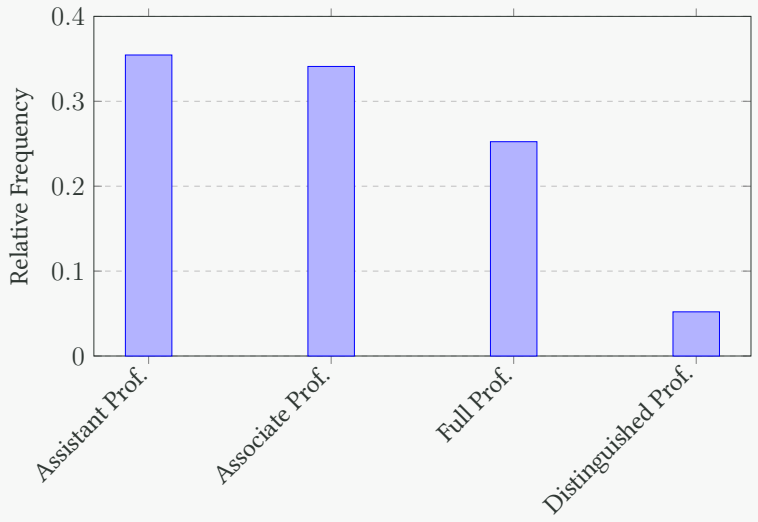
The mode is the category with the highest absolute frequency. From the table, Assistant Professor has the highest count, 184.

Answer: Assistant Professor.

Solution 5: (c) Draw a needle graph with relative frequencies

We plot the relative frequencies from part (a): one bar per rank.

Answer: Needle graph shown below.



Tip: 100% Stacked Column Chart

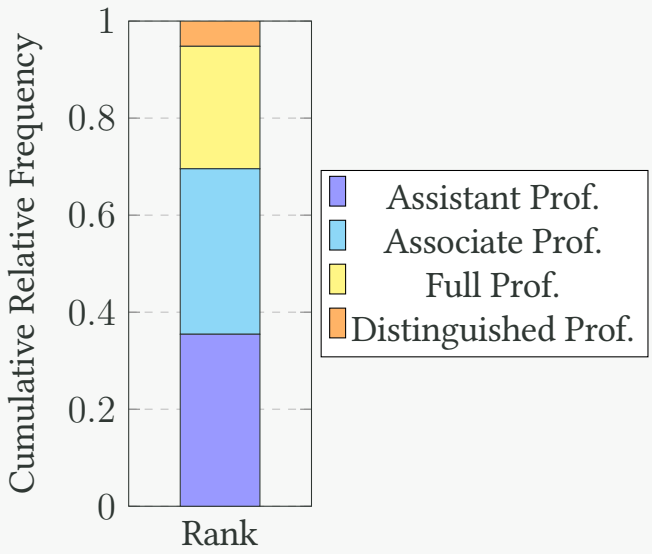
A **100% stacked column chart** shows the full sample as a single bar, split into one segment per category. Each segment's height equals that category's relative frequency, so the segment boundaries (read bottom to top) are exactly the cumulative relative frequencies, and the bar always reaches 100% at the top.

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Solution 5: (d) Draw a 100% stacked column chart

We stack the relative frequencies from part (a) in rank order into a single bar: Assistant Professor at the bottom, Distinguished Professor at the top.

Answer: Stacked column chart shown below.



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Question 6

For a population of employees, a sample was taken, and the following age distribution was observed. The first group corresponds to ages 18 to 25.

Age Group	$\hat{p}_i$
1	0.15
2	0.25
3	0.40
4	0.20

Now consider the subpopulation of employees **older than 25**. Determine the cumulative relative frequencies for the age groups 2, 3, and 4.

Tip: Rescaling Relative Frequencies within a Subpopulation

When you restrict to a subpopulation, the remaining relative frequencies no longer sum to 1. To fix this, rescale each one by dividing by the sum of the relative frequencies of the groups you kept:

$$\hat{p}_i^{\text{new}} = \frac{\hat{p}_i}{\hat{p}_{\text{first kept group}} + \cdots + \hat{p}_{\text{last kept group}}}$$

The new relative frequencies then sum to 1 again, restricted to the subpopulation.

Solution 6: Determine the cumulative relative frequencies for the age groups 2, 3, and 4, restricted to employees older than 25

Before computing anything, we set up the framework:

- **Random experiment:** Pick a random employee and record their age group.
- **Random variable:**

X	age group of a random employee

X is qualitative, with $k = 4$ categories: $m_1 = [18, 25]$, m_2, m_3, m_4 (the next three age groups). It is ordinal, since age groups have a natural order from young to old.

- **Given:** The relative frequencies $\hat{p}_i$ for each age group, as shown in the table above.

Step 1: Restrict to the subpopulation.

"Older than 25" excludes group m_1, leaving groups m_2, m_3, m_4. Within this subpopulation, the sample size is $n_2 + n_3 + n_4$.

Step 2: Rescale the relative frequencies.

We only have relative frequencies, not absolute frequencies, so we work directly with proportions. Dividing numerator and denominator of $\frac{n_i}{n_2+n_3+n_4}$ by n gives:

$$\hat{p}_i^{\text{new}} = \frac{n_i/n}{(n_2 + n_3 + n_4)/n} = \frac{\hat{p}_i}{\hat{p}_2 + \hat{p}_3 + \hat{p}_4}$$

$$\hat{p}_2^{\text{new}} = \frac{0.25}{0.85} = 0.2941, \quad \hat{p}_3^{\text{new}} = \frac{0.40}{0.85} = 0.4706, \quad \hat{p}_4^{\text{new}} = \frac{0.20}{0.85} = 0.2353$$

As a check, the new relative frequencies sum to 1:

$$0.2941 + 0.4706 + 0.2353 = 1$$

Answer:

Age Group	Rel. Freq.
2	0.2941
3	0.4706
4	0.2353

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Chapter D2 - Quantitative Variables

Chapter D3 - Relationship Between Variables

Chapter P1 - Probability

Chapter P2 - Discrete Variables

Chapter P3 - Continuous Variables

Chapter P4 - Transformations of Random Variables

Exam: 2020 June 1

Exam: 2020 June 2

Exam: 2021 June 1

Exam: 2021 June 2

Exam: 2021 August

Exam: 2022 Trial

Exam: 2022 Example

Exam: 2024 June

Exam: 2024 August

Exam: 2025 Trial

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